Student Performance - Data Cleaning & Transformation

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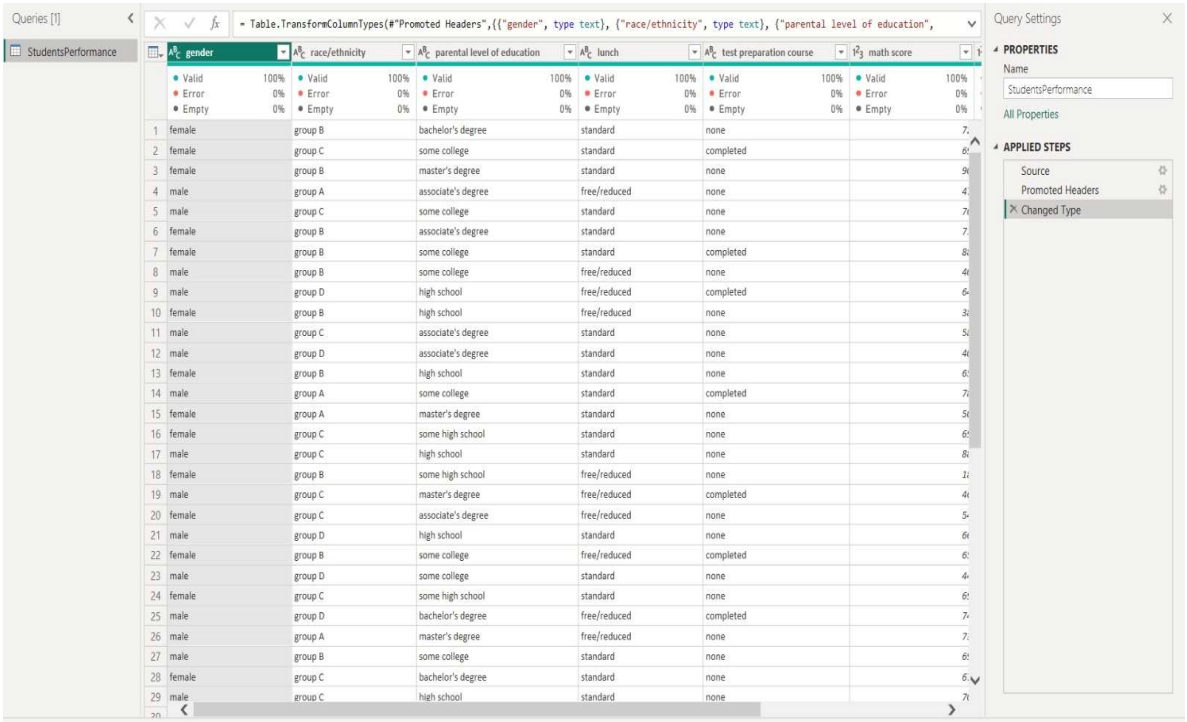
1. Executive Summary

This report outlines the step-by-step process of importing, cleaning, and transforming the StudentsPerformance dataset using Power Query Editor in Power BI Desktop. The primary objective was to derive key metrics such as Total Marks, Average, Grade, and Pass/Fail status for students.

2. Step-by-Step Data Transformation Process

Step 1: Data Import & Query Overview

The StudentsPerformance.csv dataset was successfully loaded into Power Query Editor. Initial data types were checked and auto-promoted headers were validated.

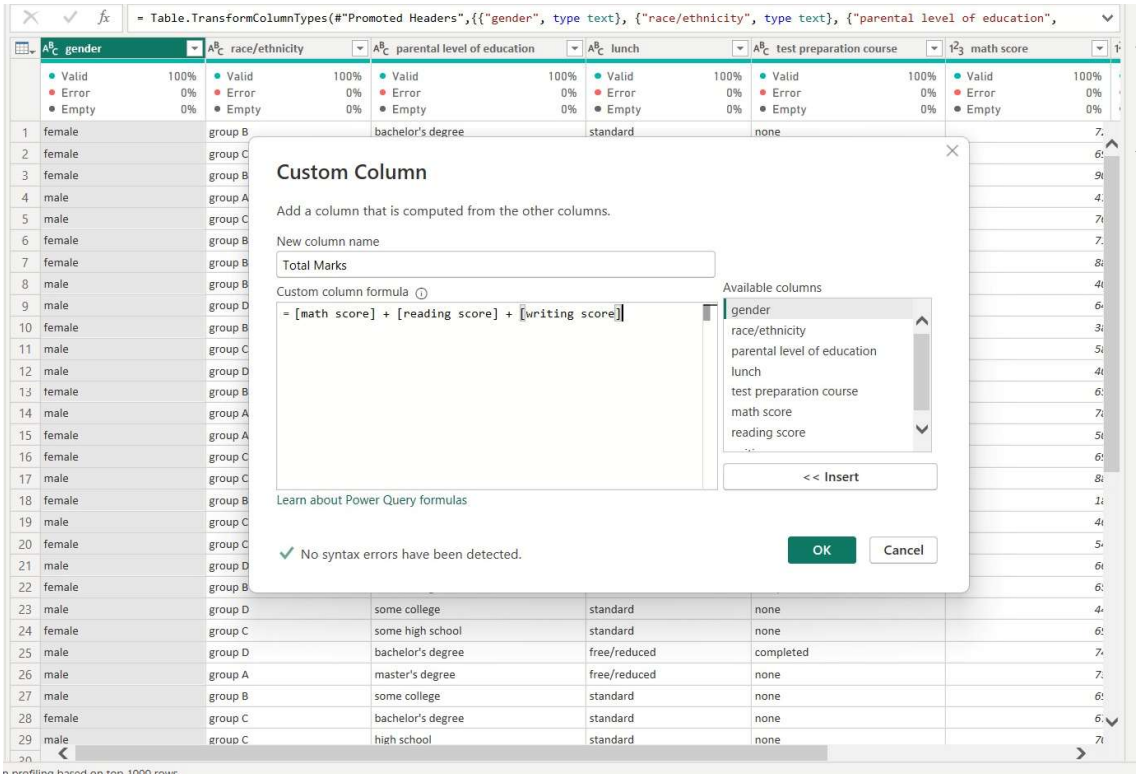


Step 2: Total Marks Calculation

A custom column named Total Marks was created to aggregate individual scores across Mathematics, Reading, and Writing subjects.

Formula Used:

[math score] + [reading score] + [writing score]

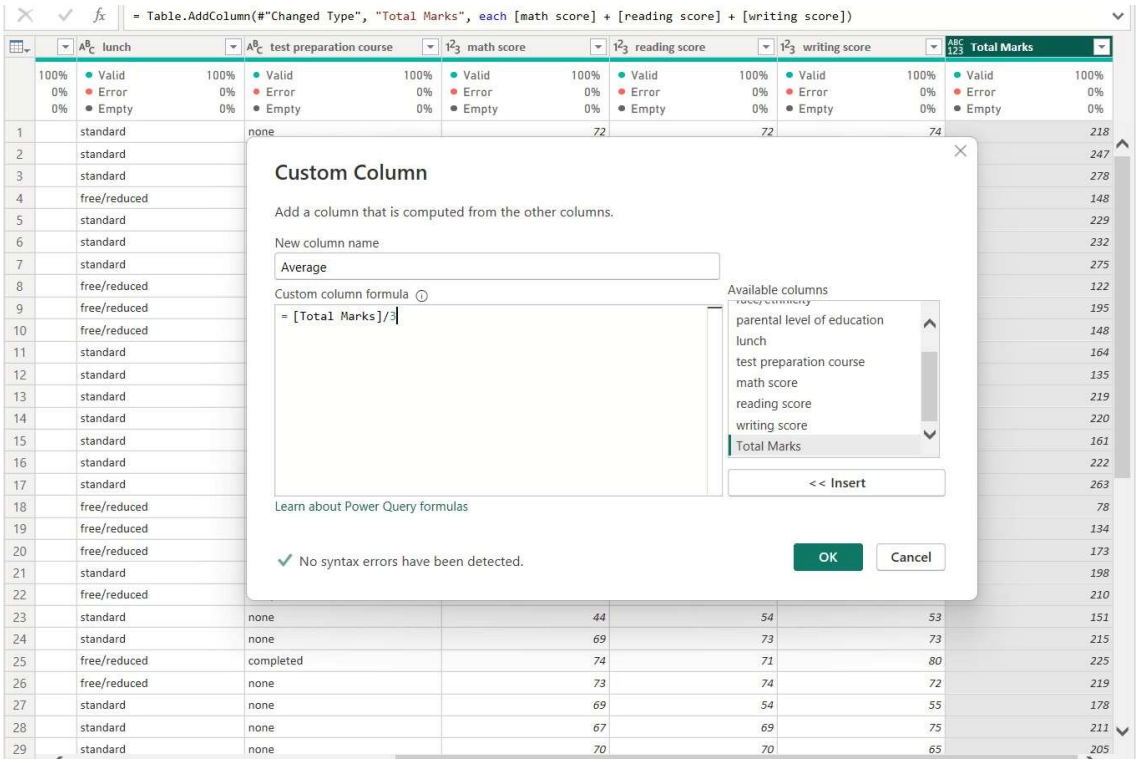


Step 3: Average Marks Calculation

To find the overall average score for each student, a second custom column named Average was added by dividing Total Marks by 3.

Formula Used:

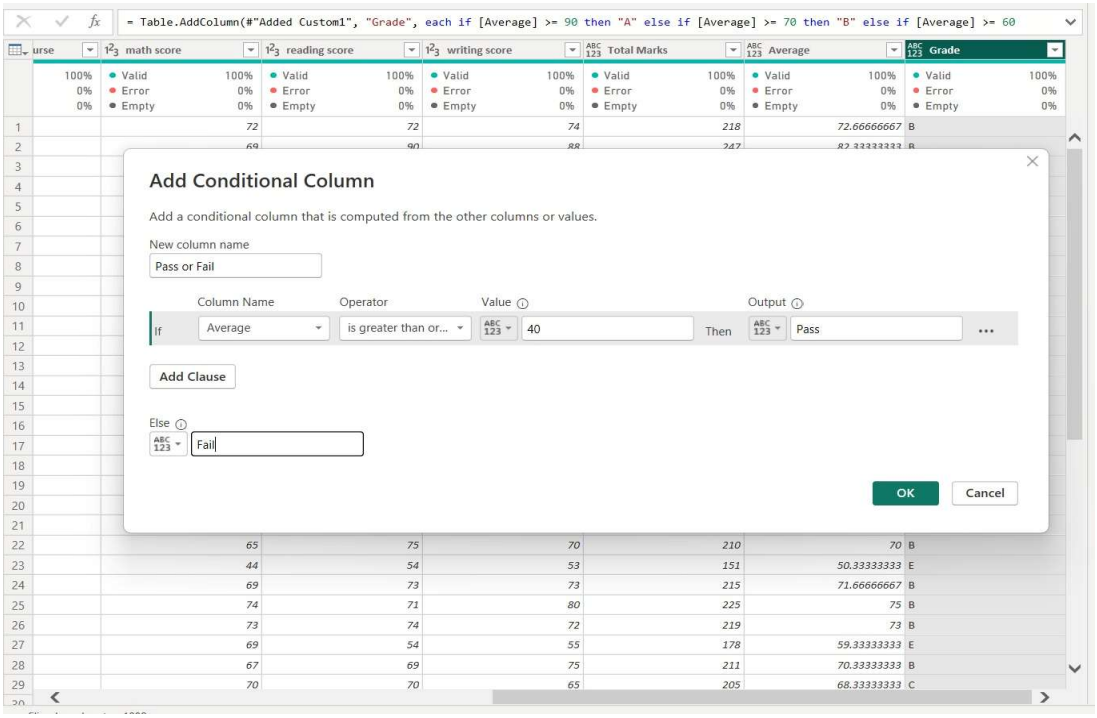
[Total Marks] / 3



Step 4: Grade Allocation via Conditional Column

Students were categorized into academic performance grades (A, B, C, E, F) based on their calculated Average scores.

- **Conditions Implemented:**
  - If Average >= 90 then A
  - Else if Average >= 70 then B
  - Else if Average >= 60 then C
  - Else if Average >= 50 then E
  - Else F

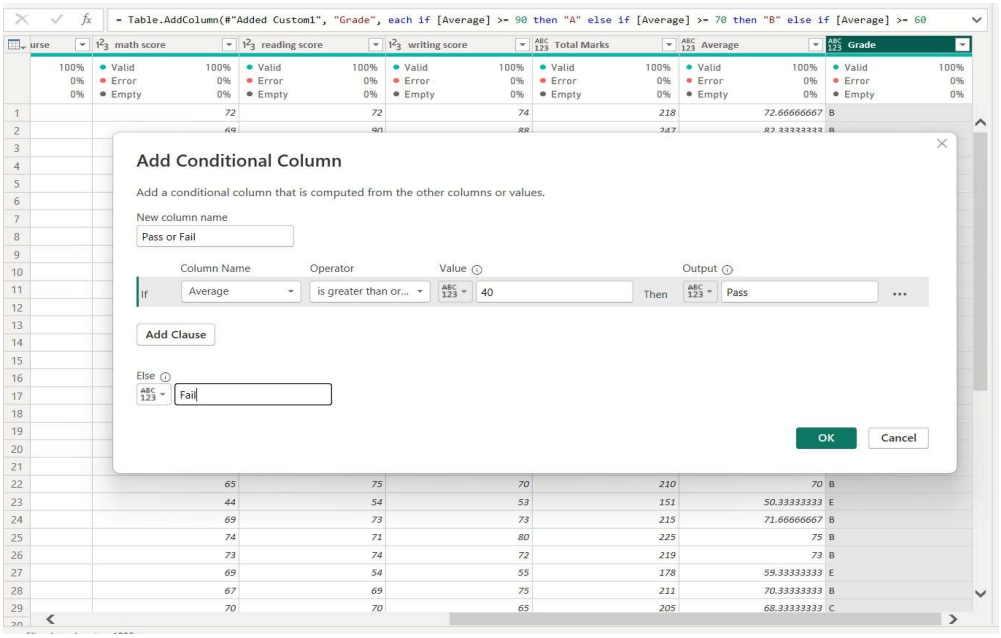


Step 5: Pass or Fail Classification

A conditional column named Pass or Fail was added to determine pass status using a benchmark threshold of 40%.

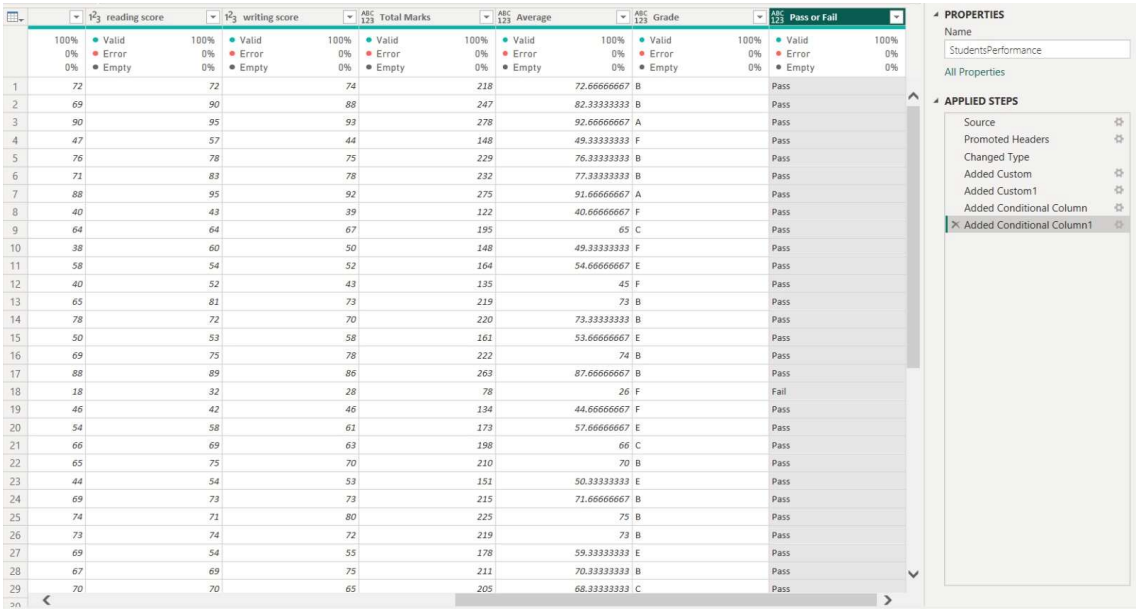
Conditions Implemented:

- If Average >= 40 then Pass
- Else Fail



Step 6: Final Transformed Dataset

All required fields (Total Marks, Average, Grade, Pass or Fail) were successfully created and added as applied steps in Power Query. The data was then saved and loaded using Close & Apply.



3. Conclusion

The dataset transformation phase was completed successfully with zero syntax errors, preparing the model for downstream DAX calculations and visual dashboard reporting.